Combinational verilog Codes

1. Write RTL for 1bit Full adder using Dataflow abstraction and verify the same using a Testbench.
2. Write RTL for 2x4 decoder using Dataflow abstraction and verify the same using a Testbench.
3. Write RTL for 8x3 priority encoder using structural model and verify the same using a Testbench.
4. Write RTL for the 4 bits Ripple carry Adder using 1-bit Full adder and verify the same using a Testbench.
5. Write RTL for 4:1 Mux using 2:1 Muxes and verify the same using a Testbench.
6. Write RTL description and test bench for 3:8 Decoder.
7. Write RTL description and testbench for 8:3 Priority encoder.

Sequential Circuits

1. Write RTL and Testbench for SR latch using Gate level modelling.
2. Write RTL and Testbench for JK Flip Flop, using parameter declaration for the respective scenarios (HOLD, TOGGLE, SET, RESET).
3. Write RTL and Testbench for for a T Flip Flop using D Flip Flop.
4. Write RTL and Testbench for a 4-bit synchronous and loadable binary up counter.
5. Write RTL and Testbench to design a 4-bit MOD-12 loadable binary synchronous up counter.
6. Write RTL and Testbench to design a 4-bit loadable binary synchronous up-down counter